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1. (10 points) The ciphertext 5859 was obtained from the RSA algorithm using $n = 11413$ and $e = 7467$. Using the factorization $11413 = 101 \cdot 113$, find the plaintext.
2. (10 points) Bob sets up a budget RSA cryptosystem. He chooses $p = 23$ and $q = 19$ and computes $n = 437 = pq$. He chooses the encryption exponent to be $e = 397$. Alice sends Bob the ciphertext $c = 123$. What is the plaintext? (You know p , q , and e .)
3. (10 points) Bob's RSA modulus is $979 = 11 \times 89$ and his encryption exponent is $e = 587$. Alice sends him the ciphertext $c = 10$. What is the plaintext?
4. (10 points) The ciphertext 62 was obtained using RSA with $n = 667$ and $e = 3$. You know that the plaintext is either 9 or 10. Determine which it is without factoring n .
5. (10 points) Alice and Bob are trying to use RSA, but Bob knows only one large prime, namely $p = 1093$. He sends $n = p = 1093$ and $e = 361$ to Alice. She encrypts her message m as $c \equiv m^e \pmod{n}$. Eve intercepts c and decrypts using the congruence $m \equiv c^d \pmod{n}$. What value of d should Eve use? Your answer should be an actual number. You may assume that Eve knows n and e , and she knows that n is prime.
6. (10 points) The exponents $e = 1$ and $e = 2$ should not be used in RSA. Why?
7. (10 points) Suppose Alice uses the RSA method as follows. She starts with a message consisting of several letters, and assigns $a = 1, b = 2, \dots, z = 26$. She then encrypts each letter separately. For example, if her message is **cat**, she calculates $3^e \pmod{n}$, $1^e \pmod{n}$, and $20^e \pmod{n}$. Then she sends the encrypted message to Bob.

Explain how Eve can find the message without factoring n . In particular, suppose $n = 8881$ and $e = 13$. Eve intercepts the message

$$4461 \quad 794 \quad 2015 \quad 2015 \quad 3603.$$

Find the message without factoring 8881.

8. (10 points) Suppose $n = pqr$ is the product of three distinct primes. How would an RSA-type scheme work in this case? In particular, what relation would e and d satisfy?

Note: There does not seem to be any advantage in using three primes instead of two. The running times of some factorization methods depend on the size of the smallest prime factor. Therefore, if three primes are used, the size of n must be increased in order to achieve the same level of security as obtained with two primes.

9. Here is a 3-person encryption scheme based on the same principles as RSA. A trusted entity chooses two large distinct primes p and q and computes $n = pq$, then chooses three integers k_1, k_2, k_3 with $k_1 k_2 k_3 \equiv 1 \pmod{(p-1)(q-1)}$. Alice, Bob, and Carla are given the following keys:

Alice: (n, k_1, k_2) , Bob: (n, k_2, k_3) , Carla: (n, k_1, k_3) .

- (a) (10 points) Alice has a message that she wants to send to both Bob and Carla. How can Alice encrypt the message so that both of them can decrypt it?
- (b) (10 points) Alice has a message that she wants to send only to Carla. How can Alice encrypt the message so that Carla can decrypt it but Bob cannot decrypt it?
10. (10 points) Suppose you have a secret, namely 5. You want to set up a system where four persons A, B, C, D are given shares of the secret in such a way that any two of them can determine the secret, but no one alone can determine the secret. Describe how this can be done. In particular, list the actual pieces of information (i.e., numbers) that you could give to each person to accomplish this.
11. (10 points) You set up a $(2, 10)$ Shamir threshold scheme, working modulo the prime 73. Recall that a share consists of a pair (a, b) , which means that the holder knows a point on the secret polynomial f , namely that $f(a) = b \pmod{73}$. Two of the shares are $(1, 10)$ and $(2, 18)$. A third share is $(5, *)$. What is $*$?
12. (10 points) In a $(3, 5)$ Shamir secret sharing scheme with modulus $p = 17$, the following shares were given to Alice, Bob, and Charles: $(1, 8)$, $(3, 10)$, $(5, 11)$, where a share (a, b) means that $f(a) = b \pmod{17}$ for the secret polynomial f . Calculate the corresponding Lagrange interpolating polynomial, and identify the secret.
13. (10 points) A certain military office consists of one general, two colonels, and five desk clerks. They have control of a powerful missile but do not want the missile launched unless the general decides to launch it, or the two colonels decide to launch it, or the five desk clerks decide to launch it, or one colonel and three desk clerks decide to launch it. Describe how you would do this with a secret sharing scheme. (*Hint: Try distributing the shares of a $(10, 30)$ Shamir scheme.*)
14. (10 points) Consider the following situation: Government A, Government B, and Government C are hostile to each other, but the common threat of Antarctica looms over them. They each send a delegation consisting of 10 members to an international summit to consider the threat that Antarctica's penguins pose to world security. They decide to keep a watchful eye on their tuxedoed rivals. However, they also decide that if the birds get too rowdy, then they will launch a full-force attack on Antarctica. Using secret sharing techniques, describe how they can arrange to share the launch codes so that it is necessary that three members from delegation A, four members from delegation B, and two members from C cooperate to reconstruct the launch codes.